

Outline

Matrices

Matrix-vector multiplication

Examples

Matrix-vector product

- ▶ *matrix-vector product* of $m \times n$ matrix A , n -vector x , denoted $y = Ax$, with

$$y_i = A_{i1}x_1 + \cdots + A_{in}x_n, \quad i = 1, \dots, m$$

- ▶ for example,

$$\begin{bmatrix} 0 & 2 & -1 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Row interpretation

- ▶ $y = Ax$ can be expressed as

$$y_i = b_i^T x, \quad i = 1, \dots, m$$

where $b_1^T, \dots, b_m^T$ are rows of A

- ▶ so $y = Ax$ is a ‘batch’ inner product of all rows of A with x
- ▶ example: $A\mathbf{1}$ is vector of row sums of matrix A

Column interpretation

- ▶ $y = Ax$ can be expressed as

$$y = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n$$

where $a_1, \dots, a_n$ are columns of A

- ▶ so $y = Ax$ is linear combination of columns of A , with coefficients $x_1, \dots, x_n$
- ▶ important example: $Ae_j = a_j$
- ▶ columns of A are linearly independent if $Ax = 0$ implies $x = 0$

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General examples

- ▶ $0x = 0$, *i.e.*, multiplying by zero matrix gives zero
- ▶ $Ix = x$, *i.e.*, multiplying by identity matrix does nothing
- ▶ inner product $a^T b$ is matrix-vector product of $1 \times n$ matrix a^T and n -vector b
- ▶ $\tilde{x} = Ax$ is de-meanned version of x , with

$$A = \begin{bmatrix} 1 - 1/n & -1/n & \cdots & -1/n \\ -1/n & 1 - 1/n & \cdots & -1/n \\ \vdots & & \ddots & \vdots \\ -1/n & -1/n & \cdots & 1 - 1/n \end{bmatrix}$$

Difference matrix

- $(n - 1) \times n$ difference matrix is

$$D = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 & 0 \\ & & \ddots & \ddots & & & \\ & & & \ddots & \ddots & & \\ 0 & 0 & 0 & \cdots & -1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

$y = Dx$ is $(n - 1)$ -vector of differences of consecutive entries of x :

$$Dx = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ \vdots \\ x_n - x_{n-1} \end{bmatrix}$$

- *Dirichlet energy*: $\|Dx\|^2$ is measure of wiggleness for x a time series

Feature matrix – weight vector

- ▶ $X = [x_1 \cdots x_N]$ is $n \times N$ *feature matrix*
- ▶ column x_j is feature n -vector for object or example j
- ▶ X_{ij} is value of feature i for example j
- ▶ n -vector w is weight vector
- ▶ $s = X^T w$ is vector of scores for each example; $s_j = x_j^T w$

7. Matrix examples

Outline

Geometric transformations

Selectors

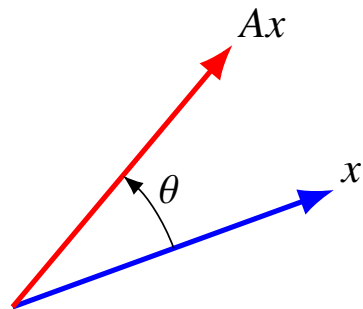
Incidence matrix

Convolution

Geometric transformations

- ▶ many geometric transformations and mappings of 2-D and 3-D vectors can be represented via matrix multiplication $y = Ax$
- ▶ for example, rotation by θ :

$$y = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} x$$



(to get the entries, look at Ae_1 and Ae_2)